

A Bayesian Joint Longitudinal–Interval and Threshold-Crossing Framework for Irregular Molecular Monitoring in Chronic Myeloid Leukemia

Abstract

Routine molecular monitoring in chronic myeloid leukemia (CML) produces data that violate the assumptions of standard survival and landmark analyses: log-transformed minimal residual disease (log-MRD) is measured at irregular clinic visits, is left-censored at an assay reporting floor, and the endpoint of deep molecular response (DMR) is detected only at the discrete visits at which a patient is tested. We develop a Bayesian joint framework that models the latent MRD trajectory and the DMR event process coherently while respecting these features. A longitudinal sub-model represents patient-specific MRD dynamics with left-censoring for assay-floor observations; an interval sub-model links the latent trajectory to first documented DMR. To resolve the circularity created by defining the endpoint through the same biomarker used as its predictor, we reformulate DMR onset as a latent threshold-crossing event and derive its interval-observed likelihood. We further develop dynamic prediction with horizon-specific recalibration and an informative visit-process extension. In a real-world cohort of 87 patients (495 MRD observations, 275 at-risk intervals), the joint model was computationally stable, calibrated at the interval level, and outperformed Kaplan–Meier, interval-only, landmark, and exact-floor alternatives by Brier score, aligning molecular-response modelling with the clinical observation process.

Keywords: joint longitudinal–survival models; interval-censored events; left-censored biomarkers; latent threshold-crossing; dynamic prediction and recalibration; Bayesian hierarchical modelling; chronic myeloid leukemia; deep molecular response.

1 Introduction

Tyrosine kinase inhibitor (TKI) therapy has transformed the prognosis of chronic myeloid leukemia (CML), shifting long-term care toward sustained molecular control, adherence, adverse-effect management, and the possible selection of patients for treatment-free remission [1–4]. Deep and durable molecular response, including MR4.5 and deep molecular response (DMR), is central to long-term response assessment and to discussions about treatment discontinuation [5–8]. Consequently, statistical models for CML are increasingly asked to turn longitudinal molecular histories into individualized, calibrated statements about response.

Routine molecular monitoring data, however, rarely have the structure assumed by simple survival or prediction models. Three features recur. First, measurements are obtained at *irregular* clinical visits, so the exact time at which a patient first attains DMR is never observed; only the interval between two visits that brackets the transition is known. Treating first *observed* DMR as an exact event time ignores this interval observation and confounds biological onset with visit timing. Second, deep molecular values are frequently reported at an assay or reporting floor. Treating floor values as exact numerical measurements understates uncertainty below the reporting limit;

the correct information is that the true burden lies at or below the floor. Third, and more subtly, the clinical endpoint is *defined* by a threshold on the same biomarker that any joint model uses as a predictor: DMR is $\log\text{-MRD} \leq -4.5$. Using the latent biomarker to predict a separately coded DMR “event” can therefore be circular.

Joint longitudinal–survival models provide a natural framework for connecting repeated biomarkers with event processes [9, 10], and recent methodological work has extended them to complex observation schemes and flexible biomarker–event associations [11, 12]. What is missing for molecular monitoring is a single framework that simultaneously (i) handles assay-floor left-censoring in the biomarker, (ii) respects the interval nature of endpoint detection under irregular visits, and (iii) removes endpoint circularity by defining the event directly through the latent biomarker trajectory, while (iv) delivering calibrated, individualized dynamic predictions without overstating clinical decision utility.

This paper develops such a framework and illustrates it on a real-world CML cohort. Our contributions are fourfold. We specify a Bayesian joint longitudinal–interval model in which patient-specific MRD trajectories with left-censored assay-floor observations drive an interval-hazard model for first documented DMR (Sections 3.1–3.2). We introduce a *latent threshold-crossing* formulation that defines DMR onset as the first time the latent MRD trajectory crosses the clinical threshold, deriving both a deterministic and a probabilistic interval-observed likelihood that resolve endpoint circularity (Section 3.3). We develop dynamic prediction with horizon-specific logistic recalibration (Section 3.4) and an informative visit-process extension that models monitoring intensity as a function of the latent trajectory (Section 3.5). Finally, we specify a simulation design to evaluate the framework under irregular visits, floor censoring, and informative monitoring (Section 3.8). The joint longitudinal–interval model is fitted to the cohort and assessed by posterior predictive checks, calibration, model comparison, and dynamic prediction (Section 4); the threshold-crossing, informative-visit, and simulation components are developed here as methodology, with their empirical evaluation and external validation the subject of ongoing work.

2 Motivating application: a CML molecular-monitoring cohort

We motivate and illustrate the methodology with a real-world CML monitoring cohort. The raw visit-level file contained 504 molecular records on 89 patients. After removing records with missing or invalid essential modelling fields, the longitudinal analysis file contained 495 complete molecular observations from 87 patients; the corresponding patient-level file contained 87 records and 275 at-risk intervals for first documented DMR (Table 1). Analysis files used coded identifiers; direct identifiers were held only in a private linkage file.

DMR was defined as $\log\text{-MRD} \leq -4.5$ and complete molecular response (CMR) as $\log\text{-MRD} \leq -5.0$; both indicators were audited by recomputation from the processed values. A structural feature of the data is central to the modelling: 239 of 495 observations (48.3%) were at or below the $\log\text{-MRD}$ floor of -5.0 , and no retained observation fell in the interval $(-5.0, -4.5]$. As a result, DMR-positive and CMR-positive counts coincide in the retained data — not through a coding error, but because all DMR-positive values were reported at the -5.0 assay floor. This directly motivates left-censored modelling of floor observations and limits the ability of the data alone to distinguish DMR from deeper response. Treatment time was recorded in months for clinical summaries and converted to years for all likelihood components; if $t^{(m)}$ denotes months since imatinib initiation, the model uses $t = t^{(m)}/12$, with time effects on the $\log(1 + t)$ scale.

Monitoring was heterogeneous, with median follow-up of 39.0 months and a median of 5 visits per patient. The median visit gap was 6.0 months; 50.3% of gaps exceeded 6 months and 20.0%

Table 1: Data cleaning audit and generated analysis datasets.

Cleaning step	Count
raw_longitudinal_rows	504
raw_longitudinal_patients	89
dropped_missing_or_invalid_core_fields	9
model_ready_longitudinal_rows	495
model_ready_longitudinal_patients	87
raw_patient_table_rows	87
usable_patient_covariate_rows	71
patient_level_analysis_rows	87
complete_core_covariate_rows	62
interval_survival_rows	275
dmr_event_patients	68
censored_patients	19

exceeded 12 months. This irregularity, together with the fact that DMR is documented only at visits, is precisely what the interval and threshold-crossing likelihoods below are designed to accommodate.

3 Statistical framework

3.1 Longitudinal sub-model with assay-floor left-censoring

Let $m_i(t)$ denote the latent biological log-MRD trajectory for patient i at treatment time t (years):

$$m_i(t) = \beta_0 + \beta_1 \log(1+t) + \beta_2 \{\log(1+t)\}^2 + b_{0i} + b_{1i} \log(1+t),$$

with independent patient-specific random effects

$$b_{0i} \sim N(0, \tau_0^2), \quad b_{1i} \sim N(0, \tau_1^2).$$

An earlier specification included a random intercept–slope correlation; because it was weakly identified in this cohort, the final model treats the random effects as independent, which improved computational stability without materially changing the fit. The observation-scale mean includes a sample-source adjustment for bone-marrow versus peripheral-blood assays:

$$\mu_{ij} = m_i(t_{ij}) + \beta_{\text{BM}} I(\text{bone marrow}_{ij}).$$

For observations above the reporting floor, $y_{ij} \sim N(\mu_{ij}, \sigma_y^2)$. For observations reported at the assay floor $c_F = -5.0$, the likelihood contribution is left-censored, carrying the information that the true value lies at or below the floor:

$$\Pr(y_{ij}^* \leq c_F) = \Phi \left\{ \frac{c_F - \mu_{ij}}{\sigma_y} \right\}.$$

3.2 Interval sub-model for first documented DMR

Because DMR is detected only at monitoring visits, the event is interval-observed. For each patient the first at-risk interval ending in a DMR-positive measurement is coded as the event interval, and preceding intervals as at risk without event; patients without documented DMR by last follow-up are right-censored. Let L_{ik} and R_{ik} be the start and end of at-risk interval k , with $\Delta_{ik} = R_{ik} - L_{ik}$ and $t_{ik}^{\text{mid}} = (L_{ik} + R_{ik})/2$. The interval hazard links the latent trajectory to DMR:

$$h_{ik}^{\text{DMR}} = \exp\{\gamma_0 + \gamma_1 \log(1 + t_{ik}^{\text{mid}}) + \gamma_2 \log(1 + \Delta_{ik}) + \alpha m_i(t_{ik}^{\text{mid}})\}, \quad p_{ik}^{\text{DMR}} = 1 - \exp\{-h_{ik}^{\text{DMR}} \Delta_{ik}\}.$$

The gap term $\gamma_2 \log(1 + \Delta_{ik})$ absorbs the mechanical dependence of documentation probability on interval length, and the shared $m_i(\cdot)$ couples the two sub-models so that lower latent MRD raises the probability of documented DMR.

3.3 Latent threshold-crossing formulation

The interval-hazard model treats DMR as an event with a hazard driven by the biomarker, but DMR is *defined* by the biomarker crossing -4.5 . To represent the endpoint directly and remove this circularity, we define DMR onset as a latent threshold-crossing time,

$$T_i^{\text{DMR}} = \inf\{t \geq 0 : m_i(t) \leq c_{\text{DMR}}\}, \quad c_{\text{DMR}} = -4.5.$$

Because molecular response is observed only at clinic visits, first DMR remains interval-observed. Writing $M_i(t) = \min_{0 \leq u \leq t} m_i(u)$ for the running minimum of the latent trajectory, a patient with documented first DMR in $(L_i, R_i]$ contributes the deterministic crossing event

$$M_i(L_i) > c_{\text{DMR}}, \quad M_i(R_i) \leq c_{\text{DMR}},$$

and a censored patient contributes $M_i(C_i) > c_{\text{DMR}}$. These hard indicators are non-smooth; for gradient-based Hamiltonian Monte Carlo they are replaced by a probabilistic threshold-crossing model that additionally accommodates uncertainty in the clinical cut-point. Letting the effective patient-level threshold be $C_i \sim N(c_{\text{DMR}}, \sigma_{\text{thr}}^2)$, the likelihood contribution for an observed DMR interval is

$$\Phi\left(\frac{M_i(L_i) - c_{\text{DMR}}}{\sigma_{\text{thr}}}\right) - \Phi\left(\frac{M_i(R_i) - c_{\text{DMR}}}{\sigma_{\text{thr}}}\right),$$

and the censoring contribution is $\Phi\{(M_i(C_i) - c_{\text{DMR}})/\sigma_{\text{thr}}\}$. As $\sigma_{\text{thr}} \rightarrow 0$ the probabilistic likelihood recovers the deterministic crossing event. This formulation makes the modelled endpoint identical to the clinical definition and handles irregular monitoring by observing the crossing only through inter-visit intervals.

3.4 Dynamic prediction and recalibration

For clinical use the framework produces individualized dynamic predictions. At landmark time s the target is

$$\Pr(T_i^{\text{DMR}} \leq s + H \mid T_i^{\text{DMR}} > s, \mathcal{H}_i(s)),$$

where $\mathcal{H}_i(s)$ is the MRD history available up to s , with landmarks $s \in \{6, 12, 18, 24\}$ months and horizons $H \in \{6, 12, 24\}$ months. Because absolute probabilities from a mechanistic joint model can be miscalibrated, we apply horizon-specific logistic recalibration,

$$\text{logit}\{p_{i,\text{cal}}(s, H)\} = a_H + b_H \text{logit}\{p_{i,\text{orig}}(s, H)\},$$

with the recalibration layer assessed by bootstrap optimism correction. Strict prospective prediction requires that patient-specific random effects be updated using only $\mathcal{H}_i(s)$; the internal analysis in Section 4 conditions on the full longitudinal record and is therefore an apparent (upper-bound) assessment of calibration, which we treat accordingly.

3.5 Informative visit-process extension

Monitoring intensity may itself depend on molecular response: a patient doing well may be seen less often, and more frequent testing yields more opportunities to document DMR. To assess sensitivity to informative visiting, we model the visit intensity jointly with the trajectory,

$$\lambda_i^V(t) = \exp\{\delta_0 + \delta_1[m_i(t) + 4.5] + \delta_2 \log(1 + t) + u_i\}, \quad u_i \sim N(0, \sigma_u^2),$$

so that visit intensity depends on the latent MRD, time, and a patient-level monitoring random effect u_i . For visit times $v_{i1}, \dots, v_{iM_i}$ in $[E_i, C_i]$ the visit-process likelihood is

$$L_i^V = \left\{ \prod_{\ell=1}^{M_i} \lambda_i^V(v_{i\ell}) \right\} \exp \left\{ - \int_{E_i}^{C_i} \lambda_i^V(s) ds \right\},$$

with the cumulative intensity evaluated by quadrature. Because this extension adds weakly identified parameters in a cohort of 87 patients, it is presented as a sensitivity model to be evaluated primarily by simulation and in larger cohorts.

3.6 Priors, estimation, and computation

Estimation is fully Bayesian via Hamiltonian Monte Carlo in Stan. Weakly informative priors were fixed before fitting:

$$\beta_0 \sim N(-2.5, 2^2), \quad \beta_1 \sim N(-1, 1^2), \quad \beta_2 \sim N(0, 0.5^2), \quad \beta_{\text{BM}} \sim N(0, 1^2),$$

$$\sigma_y \sim \text{Exponential}(1), \quad \tau_0, \tau_1 \sim \text{Exponential}(1), \quad \gamma_0 \sim N(-2, 2^2), \quad \gamma_1, \gamma_2 \sim N(0, 1^2), \quad \alpha \sim N(-0.5, 0.75^2).$$

Convergence was assessed by $\hat{R}$, bulk and tail effective sample size, divergent-transition counts, tree depth, and E-BFMI. In the latent threshold-crossing and informative-visit models, the running-minimum and hard-indicator terms are implemented through smooth approximations to preserve differentiability.

3.7 Model assessment and comparison

Model adequacy was evaluated with posterior predictive checks for the longitudinal component (predictive coverage; observed versus predicted assay-floor rate) and with calibration of DMR probabilities at both the interval and patient level (calibration-in-the-large, calibration slope, and grouped calibration plots). Discrimination and overall accuracy were summarized by Brier scores. The joint model was compared against four simpler alternatives that a practitioner might otherwise use: a descriptive Kaplan–Meier analysis of first observed DMR; an interval-only model with timing and visit-gap terms; a landmark MRD model; and a joint model that treats floor observations as exact -5.0 . Refittable simpler models were internally validated by patient-cluster bootstrap with optimism correction; for the Bayesian joint models, fixed-posterior bootstrap summaries are reported.

3.8 Simulation design

To characterize the operating properties of the framework beyond the single cohort, we specify a simulation study evaluating whether the interval-hazard model, the deterministic and probabilistic threshold-crossing models, and the informative visit-process extension recover DMR onset and calibrated DMR probabilities. Data-generating scenarios cross non-informative versus informative monitoring, exact versus floor-censored molecular values, threshold uncertainty around -4.5 , sparse versus dense monitoring, and sample sizes of 87, 150, and 300 patients; performance is measured by bias, coverage, calibration, and Brier score. This design fixes the evaluation protocol used in the ongoing extension of this work.

4 Application to the CML cohort

We fit the joint longitudinal–interval model of Sections 3.1–3.2 to the cohort and report the fitted results below; the threshold-crossing, informative-visit, and simulation components are developed as methodology (Sections 3.3, 3.5, 3.8) and evaluated in ongoing work.

4.1 Cohort and endpoint summary

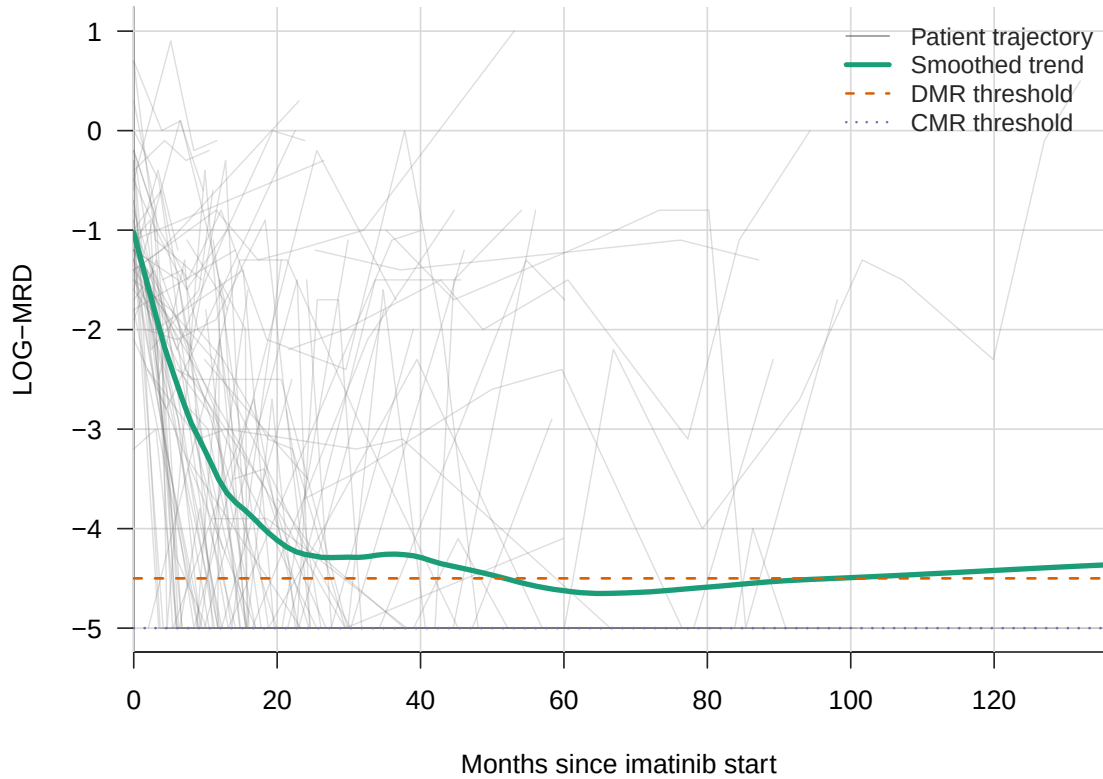
The final cohort comprised 87 patients, 495 longitudinal log-MRD observations, and 275 at-risk intervals; 68 patients had documented DMR and 19 were censored. Baseline clinical covariates were complete for 62 patients (71.3%) (Table 2). Molecular response deepened over follow-up, and

Table 2: Baseline characteristics and follow-up summaries.

Characteristic	All	DMR	Censored
Patients, n	87	68	19
Age, mean (SD)	44.6 (15.5); n=62	44.9 (15.5); n=55	42.7 (16.2); n=7
Male sex, n (%)	38 (61.3%); n=62	35 (63.6%); n=55	3 (42.9%); n=7
Imatinib duration, years, median [IQR]	5.0 [2.1, 7.5]; n=62	5.0 [3.2, 7.8]; n=55	2.0 [1.5, 3.8]; n=7
Baseline PH+ %, median [IQR]	100.0 [100.0, 100.0]; n=62	100.0 [100.0, 100.0]; n=55	100.0 [100.0, 100.0]; n=7
3-month PH+ %, median [IQR]	0.0 [0.0, 10.1]; n=44	0.0 [0.0, 10.4]; n=38	0.0 [0.0, 3.8]; n=6
6-month PH+ %, median [IQR]	0.0 [0.0, 0.0]; n=52	0.0 [0.0, 0.0]; n=45	0.0 [0.0, 7.5]; n=7
12-month PH+ %, median [IQR]	0.0 [0.0, 0.0]; n=59	0.0 [0.0, 0.0]; n=52	0.0 [0.0, 0.0]; n=7
Follow-up, months, median [IQR]	39.0 [15.8, 65.8]; n=87	46.5 [23.1, 71.7]; n=68	13.0 [10.8, 24.1]; n=19
Visits per patient, median [IQR]	5 [4, 7]; n=87	6 [4, 8]; n=68	4 [3, 4]; n=19
DMR observed, n (%)	68 (78.2%); n=87	68 (100.0%); n=68	0 (0.0%); n=19

Table 3: Longitudinal molecular monitoring summaries by follow-up window.

Window_months	Observations	Patients	BM_samples	LOG_MRD	DMR	CMR
0-3	67	55	67 (100.0%); n=67	-1.2 [-1.6, -0.8]; n=67	2 (3.0%); n=67	2 (3.0%); n=67
>3-6	36	34	36 (100.0%); n=36	-2.0 [-3.5, -1.1]; n=36	8 (22.2%); n=36	8 (22.2%); n=36
>6-12	79	59	79 (100.0%); n=79	-2.7 [-5.0, -1.6]; n=79	26 (32.9%); n=79	26 (32.9%); n=79
>12-24	95	59	92 (96.8%); n=95	-5.0 [-5.0, -2.4]; n=95	51 (53.7%); n=95	51 (53.7%); n=95
>24-60	135	53	134 (99.3%); n=135	-5.0 [-5.0, -2.4]; n=135	90 (66.7%); n=135	90 (66.7%); n=135
>60	83	26	83 (100.0%); n=83	-5.0 [-5.0, -4.5]; n=83	62 (74.7%); n=83	62 (74.7%); n=83

**Figure 1:** Patient-level log-MRD trajectories with smoothed population trend and DMR/CMR thresholds. Lines and thresholds are distinguishable in grayscale.

the assay-floor share rose with time (Table 3), consistent with the trajectory and floor-censoring structure in Figure 1.

The descriptive Kaplan–Meier curve for first observed DMR (Figure 2) is retained only as a summary, because its event times correspond to visits at which DMR was first documented and should not be read as exact biological onset.

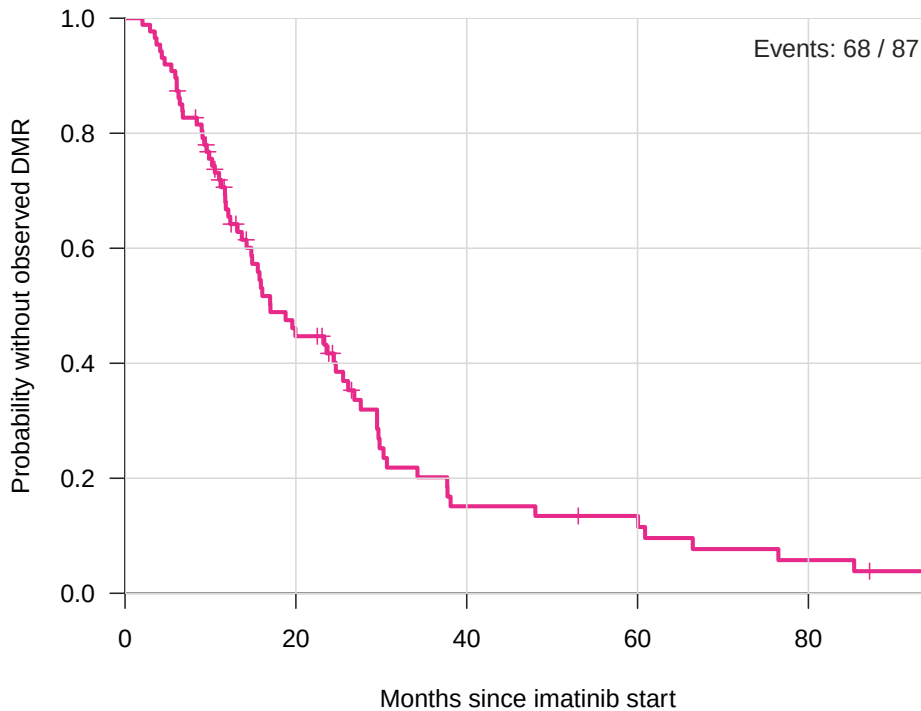


Figure 2: Descriptive Kaplan–Meier curve for time until first observed DMR. Event times correspond to monitoring visits and should not be interpreted as exact biological onset.

4.2 Fitted joint model

Computation was stable: no divergent transitions, maximum tree depth 7, minimum approximate E-BFMI 0.610, maximum $\hat{R}$ 1.003, minimum bulk effective sample size 1359, and minimum tail effective sample size 1395; no parameter had $\hat{R} > 1.01$ or effective sample size below 400. Posterior summaries are given in Table 4. The latent time effect was strongly negative ($\beta_{\text{time}} = -3.561$; 5th–95th posterior percentiles -4.351 to -2.778), indicating marked decline in latent log-MRD over follow-up, with a positive quadratic log-time term ($\beta_{\text{time}^2} = 0.502$; 0.078 to 0.914) consistent with nonlinear rather than linear dynamics. The association parameter was negative ($\alpha_{\text{MRD}} = -1.275$; -1.724 to -0.899), so lower latent MRD was associated with higher interval probability of documented DMR. These are monitoring-oriented associations, not externally validated treatment-decision thresholds.

4.3 Posterior predictive checks and calibration

Among non-floor observations, 90% posterior predictive coverage was 0.957 and 95% coverage was 0.984; the observed assay-floor rate (0.483) was close to the posterior mean floor probability (0.414)

Table 4: Renewed primary Bayesian joint-model posterior summaries. Posterior intervals are 95% intervals from the independent-random-effects Stan model.

Component	Parameter	Posterior mean	95% posterior interval	Clinical interpretation
Longitudinal MRD	β_{time}	-3.561	-4.500 to -2.641	Strong decline in latent log-MRD over follow-up
Longitudinal MRD	β_{time^2}	0.502	-0.002 to 0.999	Evidence of nonlinear response dynamics
Longitudinal MRD	β_{BM}	0.607	-0.805 to 2.031	Sample-source adjustment; uncertain effect
Longitudinal MRD	σ_y	1.813	1.634 to 2.012	Residual molecular-measurement variability
Interval DMR	γ_{time}	-0.228	-1.138 to 0.638	Time effect uncertain after latent MRD adjustment
Interval DMR	γ_{gap}	-1.831	-2.879 to -0.804	Visit-gap term; interpret with cumulative interval length
Joint association	α_{MRD}	-1.275	-1.848 to -0.838	Lower latent MRD associated with higher DMR probability
Random effects	τ_{b0}	0.968	0.510 to 1.405	Patient heterogeneity in baseline latent MRD
Random effects	τ_{b1}	2.203	1.561 to 2.948	Patient heterogeneity in response slope

(Table 5, Figure 3). Interval-level calibration was close in aggregate: observed interval event rate 0.247 versus mean predicted 0.247, Brier score 0.082. Patient-level cumulative DMR probability was underpredicted (observed 0.782 versus mean predicted 0.581; Figure 4), which motivated the dynamic prediction and recalibration analysis below.

Table 5: Diagnostics, posterior predictive checks, calibration, and sensitivity summaries for the renewed independent-random-effects joint model.

Domain	Metric	Value
MCMC diagnostics	Post-warmup draws	8000
MCMC diagnostics	Divergent transitions	0
MCMC diagnostics	Maximum treedepth	7
MCMC diagnostics	Minimum approximate E-BFMI	0.610
MCMC diagnostics	Maximum R-hat	1.003
MCMC diagnostics	Minimum bulk ESS	1359.258
MCMC diagnostics	Minimum tail ESS	1395.463
Longitudinal PPC	Non-floor RMSE to posterior mean	1.494
Longitudinal PPC	Non-floor 90% predictive coverage	0.957
Longitudinal PPC	Non-floor 95% predictive coverage	0.984
Assay-floor PPC	Observed floor rate	0.483
Assay-floor PPC	Posterior mean floor probability	0.414
Interval calibration	Observed event rate	0.247
Interval calibration	Mean predicted probability	0.247
Interval calibration	Brier score	0.082
Patient calibration	Observed DMR rate	0.782
Patient calibration	Mean predicted DMR probability	0.581
Patient calibration	Brier score	0.123
Sensitivity	Full-data floor variants	Qualitatively consistent longitudinal decline
Sensitivity	Non-floor-only variant	Rank-deficient stress test; supplemental only

4.4 Model comparison

The joint model had lower interval-level and patient-level Brier scores than the descriptive Kaplan–Meier, interval-only, landmark, and exact-floor alternatives (Table 6). The improvement over the exact-floor joint model isolates the contribution of left-censored assay-floor handling, and the improvement over the interval-only and landmark models isolates the value of coupling the longitudinal and event processes. This supports the joint framework for monitoring-oriented inference; it does

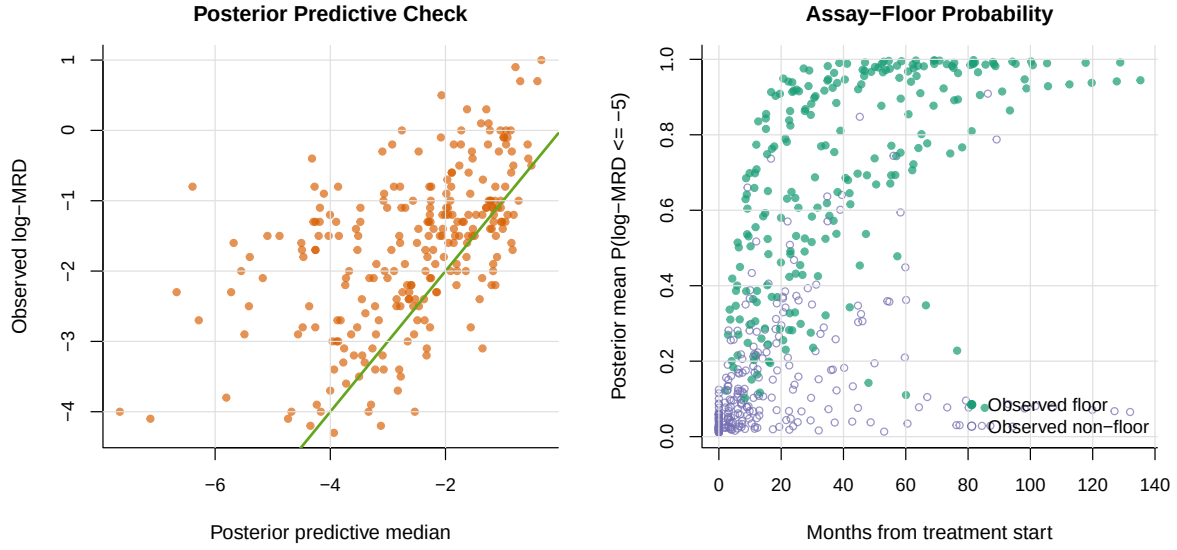


Figure 3: Posterior predictive check for longitudinal log-MRD (left) and posterior mean probability of assay-floor response (right); points distinguish observed floor and non-floor measurements.

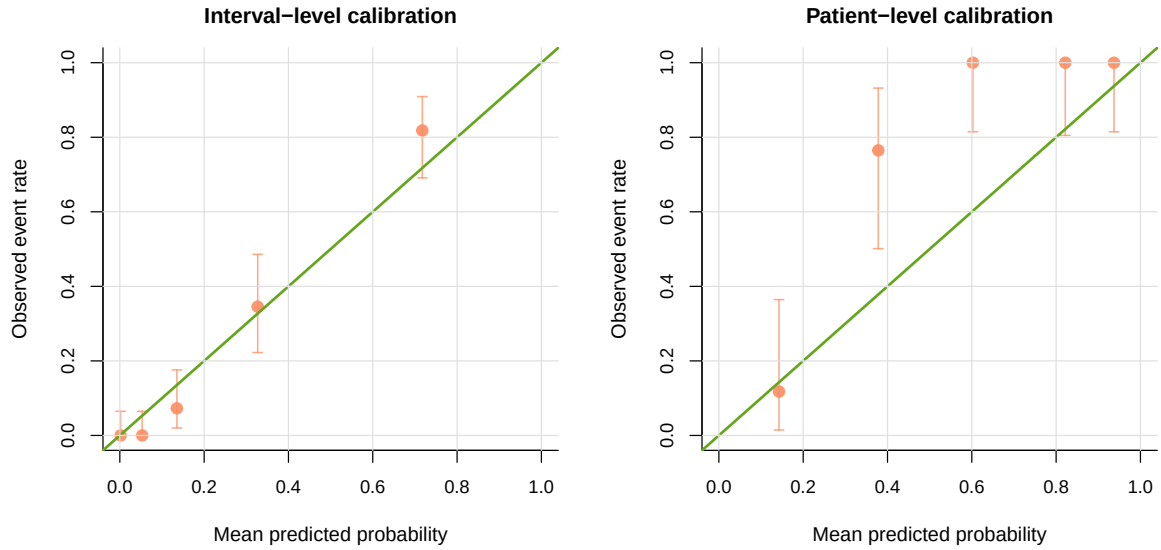


Figure 4: Grouped calibration of model-estimated DMR probabilities. Interval-level calibration is close in aggregate, whereas patient-level cumulative DMR probability is underpredicted.

not by itself establish clinical decision readiness.

Table 6: Model comparison and internal validation. Event counts are shown as evaluated records/events. Validated Brier scores are patient-cluster bootstrap optimism-corrected for refittable simpler models; for the two Stan joint models, they are fixed-posterior bootstrap summaries without Stan refitting.

Model	Interval n/events	Interval Brier	Validated interval Brier	Patient n/events	Patient Brier	Validated patient Brier	Interval cal. int./slope	Patient cal. int./slope
Descriptive Kaplan–Meier	275/68	0.172	0.182	87/68	0.362	0.370	0.497/0.433	2.059/-0.124
Interval timing + visit gap	275/68	0.174	0.178	87/68	0.290	0.296	0.000/1.000	1.271/-0.345
Landmark MRD	55/20	0.211	0.246	32/20	0.267	0.303	-0.000/1.000	0.756/0.387
Joint model, floor exact	275/68	0.094	0.094	87/68	0.123	0.123	-0.008/1.617	1.360/2.419
Primary joint model	275/68	0.082	0.082	87/68	0.116	0.116	0.003/1.784	1.329/2.816

4.5 Dynamic prediction and recalibration

The internal dynamic-prediction analysis generated 300 eligible landmark–horizon records. Before recalibration, the joint model overpredicted 6-month DMR probability (mean predicted 0.506 versus observed 0.361; Brier 0.179), modestly overpredicted the 12-month horizon (0.626 versus 0.576; Brier 0.162), and underpredicted the 24-month horizon (0.721 versus 0.828; Brier 0.081). Horizon-specific logistic recalibration aligned mean predicted probabilities with observed rates; bootstrap optimism-corrected recalibrated Brier scores were 0.167, 0.172, and 0.063 at 6, 12, and 24 months (Table 7, Figure 5). Because these predictions used patient-specific random effects estimated from the full longitudinal record, they are apparent performance; strict prospective prediction requires posterior updating restricted to the history available at each landmark.

Table 7: Development-cohort dynamic prediction and recalibration performance. Predictions are landmark-level probabilities of documented DMR by the stated horizon among patients who were DMR-free at the landmark and had at least one prior MRD measurement. Optimism correction applies to the recalibration layer only; the Bayesian joint model itself was not refitted in bootstrap samples.

Horizon	Model	n	Events	Observed	Mean predicted	Brier	Opt.-corrected Brier	Cal. intercept	Cal. slope
6 months	Original posterior	108	39	0.361	0.506	0.179	0.179	-1.00	0.85
6 months	Recalibrated	108	39	0.361	0.361	0.157	0.167	-0.00	1.00
12 months	Original posterior	99	57	0.576	0.626	0.162	0.162	-0.36	0.81
12 months	Recalibrated	99	57	0.576	0.576	0.161	0.172	0.00	1.00
24 months	Original posterior	93	77	0.828	0.721	0.081	0.081	1.02	1.81
24 months	Recalibrated	93	77	0.828	0.828	0.055	0.063	0.00	1.00

5 Discussion

5.1 Principal findings

We have developed a Bayesian framework that aligns the statistical model with the structure of routine CML molecular monitoring: repeated MRD measurements, assay-floor reporting, irregular visits, and interval-observed DMR. In the motivating cohort, the fitted joint longitudinal–interval model was computationally stable, close to calibrated at the interval level, and outperformed Kaplan–Meier, interval-only, landmark, and exact-floor alternatives by Brier score. The comparison against the exact-floor joint model shows that treating deep values as left-censored, rather than as exact -5.0 measurements, measurably improves fit.

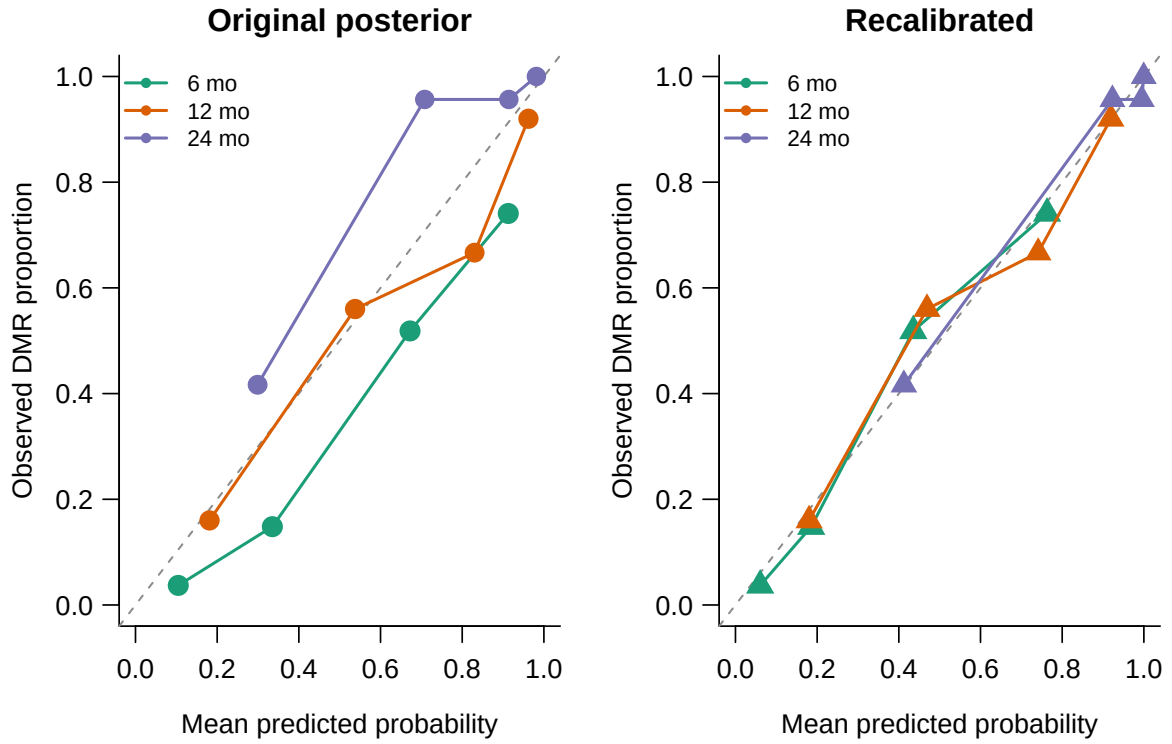


Figure 5: Dynamic DMR calibration before and after horizon-specific logistic recalibration. Points are quantile-based prediction bins, sized by bin count; the diagonal indicates perfect calibration.

5.2 Methodological contribution

The central methodological contribution is the latent threshold-crossing formulation. Because DMR is defined by $\log\text{-MRD} \leq -4.5$, using the latent trajectory to predict a separately coded DMR event is partly circular; defining onset as the first crossing of the threshold by the latent trajectory makes the modelled and clinical endpoints identical, and its interval-observed likelihood — deterministic, or probabilistic with threshold uncertainty σ_{thr} — handles irregular monitoring naturally. The probabilistic version is also better suited to gradient-based sampling than the non-smooth deterministic indicators. The identical DMR and CMR counts in the retained data are a consequence of assay-floor reporting rather than a coding artifact, and reinforce the need for left-censored modelling of floor observations.

5.3 Clinical interpretation

The fitted model should be read as a monitoring-support framework, not a treatment-decision rule. The negative association between latent MRD and interval DMR probability is biologically coherent and supports using serial molecular history for response assessment, but it does not validate the model for TKI discontinuation, treatment-free-remission eligibility, or changes in treatment. Dynamic prediction with recalibration improves calibration-in-the-large and provides individualized horizon probabilities, but the current estimates are apparent rather than strictly prospective.

5.4 Informative monitoring

Monitoring is irregular and may be informative: testing frequency can depend on molecular response, adherence, clinician concern, or institutional practice, and more frequent testing creates more opportunities to document DMR. The visit-process extension models visit intensity as a function of the latent trajectory, time, and a patient-level monitoring effect, providing a route to assess this sensitivity; because it adds weakly identified parameters in a modest cohort, it is best evaluated by simulation and in larger data.

5.5 Limitations

The cohort was modest (87 patients; complete core covariates in 62) and came from a single real-world source without an external validation cohort. Monitoring was irregular, so patients with longer or denser follow-up had more opportunity to document DMR. The retained data contained no observations in $(-5.0, -4.5]$, limiting the ability to distinguish DMR from CMR descriptively. The fitted interval-hazard model does not by itself remove endpoint circularity; the threshold-crossing formulation addresses it and is evaluated in ongoing work. The dynamic predictions used random effects from the full record and are therefore apparent. The informative-visit and simulation components are specified here as methodology; their empirical operating characteristics are established in the simulation study of Section 3.8 as that work completes.

5.6 Future work

Future work will fit and validate the deterministic and probabilistic threshold-crossing models, compare them with the interval-hazard model, and quantify whether the threshold-crossing endpoint improves calibration and interpretability; execute the simulation study of Section 3.8; implement strict prospective landmark prediction with history-restricted posterior updating; and test whether recalibrated dynamic probabilities transport to independent CML cohorts through external or temporal validation.

6 Conclusions

We have presented a Bayesian framework for CML molecular monitoring that is aligned with irregular clinical observation, assay-floor reporting, and interval-observed DMR. The fitted joint model is computationally stable and supports monitoring-oriented inference, and the latent threshold-crossing formulation resolves endpoint circularity by defining the event directly through the latent biomarker trajectory. Dynamic prediction with recalibration yields individualized horizon probabilities. Clinical decision use requires prospective calibration and external validation.

Statements and Declarations

Ethical considerations

The study used de-identified, retrospectively collected molecular monitoring data. The relevant Institutional Review Board / Ethics Committee approval or waiver (name, institution, and approval number) is provided on the Title Page and will be stated here in the non-anonymized version.

Consent to participate

As detailed on the Title Page, the requirement for informed consent was addressed under the approving committee’s determination for retrospective, de-identified data.

Consent for publication

Not applicable; the manuscript reports de-identified aggregate analyses and contains no individually identifying data.

Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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Data availability

The analysis workflow generated privacy-preserving processed datasets, model code, tables, and figures. Direct identifiers are excluded from all analysis files and retained only in a private linkage file; processed data and code supporting the findings are available from the corresponding author on reasonable request, subject to institutional and ethical approval.

Reporting guidelines

Development and reporting of the prediction model followed the TRIPOD recommendations; the completed checklist is provided as a supplementary file.

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